Band-1 Hadamard Order 2308 — Miyamoto (H_576 Kronecker) + Paley-II Cross-Check; Gap 2052 Triage

H_24/H_576/C_578/H_2308 PASS (Miyamoto + Paley-II); 2308 in SPK via Paley II q=1153; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_2276_Miyamoto`; `2026-09-14_Band1_Hadamard_next_gap_after_2036_`
Lab. `/workspace/experiments/band1/hadamard_2308_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 24, 576, and 2308 only** (verified max $|HH^\top - nI| = 0$), via **two independent paths** for order 2308 (Miyamoto primary; Paley type II cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure; method triage below is a **non-finding only** (non-finding ≠ counterexample). Order 2308 **enters SPK** via Paley II ($q = 1153$); it is **not** a Miyamoto-only bonus (contrast prior 2276). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders 2116 / 2164 / 2228 / 2276 on box. This increment constructs the **next** Miyamoto-eligible target after $q = 569$: **order 2308** ($2308 = 4 \cdot 577$, 577 prime $\equiv 1 \pmod 4$). Direct Paley-I/II for H_{576} **fail** (575 and 287 composite). On box: H_{24} via Paley type I ($q = 23$), $H_{576} = \text{Kronecker}(H_{24}, H_{24})$ (alt cross-check Kronecker(H_{16}, H_{36}) also PASS), symmetric Paley conference C_{578} over prime 577, and $H_{2308} = \text{Miyamoto}(H_{576}, C_{578})$. Independently, H_{2308} also verifies via **Paley type II** with $q = 1153$ (explains SPK membership). Unit tests **23/23 PASS**. SPK next gap remains **2052** (OPEN; method triage documents Paley-I/II / Kronecker / Miyamoto-Paley-conf FAIL). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2308; Miyamoto; Paley type II; Kronecker; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{24} constructed and verified (Paley type I, $q = 23$). 2. H_{576} constructed and verified (Kronecker $H_{24} \otimes H_{24}$). Direct Paley-I/II paths **rejected**. Alt Kronecker $H_{16} \otimes H_{36}$ also PASS. 3. C_{578} constructed and verified (Paley conference, $q = 577$): $CC^\top = 577I$, zero diagonal. 4. H_{2308} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type II ($q = 1153$). **Not** the Hadamard conjecture. 5. SPK: $2308 \in$ SPK (via Paley II). Next constructive gap remains **2052** (OPEN). Method triage for 2052 is non-finding only.

2. Arithmetic and eligibility

Quantity	Value
Target order	$2308 = 4 \cdot 577$
$q = 577$	prime, $577 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{576}
Needed conference	order $578 = q + 1$
H_{576} path (used)	Paley I $q = 23 \rightarrow H_{24}$; Kronecker $H_{24} \otimes H_{24}$
H_{576} Paley-I direct	**REJECTED**: $q' = 575$ not prime
H_{576} Paley-II direct	**REJECTED**: $q' = 287 = 7 \cdot 41$ not prime
H_{576} alt	Kronecker(H_{16} Sylvester, H_{36} Paley-II $q = 17$)
Conference path	Symmetric Paley over prime 577
SPK path for 2308	Paley type II, $q = 1153$ prime $\equiv 1 \pmod 4$; order $2(q + 1) = 2308$

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{24}	Paley type I, $q = 23$	0	0.0002	**PASS**
H_{576}	Kronecker(H_{24}, H_{24})	0	0.0865	**PASS**
H_{576} alt	Kronecker(H_{16}, H_{36})	0	0.0839	**PASS**
C_{578}	Paley conference, $q = 577$	0	0.2193	**PASS**
H_{2308}	Miyamoto(H_{576}, C_{578})	0	5.0286	**PASS**
H_{2308} P-II	Paley type II, $q = 1153$	0	5.7657	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh, excl. separate Paley-II cross-check wall) ≈ 5.62 s. Compressed ± 1 matrices saved as `H24_pm1.npz`, `H576_pm1.npz`, `H2308_pm1.npz`, `H2308_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\{508, 528, 540, 556, 568, 284, 2004, 2036, 2116, 2164, 2228, 2276\}$ $\cup$ verified H_{24} / $H_{576} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2308 enters SPK **because** Paley II applies ($q = 1153$).

Quantity	Value
Known count	880
2308 in SPK	**True** (Paley II $q = 1153$)
576 in SPK	True (Kronecker 24×24)
2276 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 method triage (non-finding ONLY)

Method	Result	Reason
Paley I	**FAIL**	$q = 2051$ not prime (or not $\equiv 3 \pmod 4$ as required)
Paley II	**FAIL**	$q = 1025 = 25 \cdot 41$ not prime
Kronecker	**FAIL**	$2052 = 2^2 \cdot 3^3 \cdot 19$; **no** Hadamard-admissible factor split

Method	Result	Reason
Miyamoto + Paley conf	**FAIL**	$q = 513 = 3^3 \cdot 19$ not prime; Paley conference needs prime $q \equiv 1 \pmod 4$
Goethals–Seidel / SDS ~513	lit-only this run	No SDS constructed on box
Williamson / Turyn / Yamada-A	lit-only this run	Not constructed on box

Claim level. Method triage / non-finding ONLY. Gap **2052 remains OPEN**. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (23/23)

Test	Result
`H20_paley_I_control`	PASS
`H4_sylvester`	PASS
`paley_II_q5_H12`	PASS
`paley_II_q13_H28`	PASS
`miyamoto_H20_from_H4_conf6`	PASS
`paley_I_q23_H24`	PASS
`paley_II_q17_H36`	PASS
`kronecker_H4xH4_H16`	PASS
`q577_is_prime` / `q577_equiv_1_mod_4` / `n2308_equals_4_times_577`	PASS
`q23` / `H24` / `H576` / `C578` formulae	PASS
`paley_I_H576_path_rejected`	PASS
`paley_II_H576_path_rejected`	PASS
`reject_all_ones_H8`	PASS
`conference_6_paley_control`	PASS
`miyamoto_eligible_2308_arith`	PASS
`conference_578_shape_expected`	PASS
`gap_2052_not_filled_by_this_ship`	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 577$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
593	2372	592	Kronecker(4×148)
601	2404	600	**Paley I** ($q = 599$)
613	2452	612	no quick SPK seed noted
617	2468	616	Double(Paley I half 308)
641	2564	640	Kronecker(4×160)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct H_{2404} via Paley-I H_{600} + Miyamoto, **or** Track B / Object 1 await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Full Hadamard conjecture: **OPEN**
- Clay-closed / finished-ToE: **false / not claimed**
- Gap 2052 filled: **NOT claimed** (OPEN)
- GATE-04: **LOCKED**
- RH / P≠NP / YM / Frankl / NS-solved / AC-solved: **not claimed**

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2308_miyamoto
python3 verify_hadamard_2308.py
# expects: Unit tests 23/23; H24/H576/C578/H2308 Miyamoto PASS; Paley-II cross-check PASS
# artifacts: results.json, literature_ledger.json, SUMMARY.txt, *.npz
```

Lab path. `/workspace/experiments/band1/hadamard_2308_miyamoto/`
PDF shelf. `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2308_Miyamoto.pdf`